Léo Crampé

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Education

University of Southern California **Los Angeles, CA**
Master of Science in Mechanical Engineering (Artificial Intelligence and Machine Learning Emphasis) December 2027
Bachelor of Science in Biomedical Engineering (Mechanical Emphasis): **3.66 GPA** May 2027
Relevant Coursework: Simulation Methods, Rehabilitation Engineering, Applications of AI/ML, Mechanics of Materials

Anglo-American University **Prague, Czechia**
Study Abroad Program August 2023 - December 2023

Experience

NeuroSense **Los Angeles, CA**
Co-Founder September 2025 - September 2026
- Engineered a portable vibration threshold testing device utilizing **Arduino**, linear actuating motors, **Solidworks**, and **3D printing** to objectively measure sensory loss in patients with peripheral neuropathy
- Advanced NeuroSense to the USC Viterbi Min Family Challenge **semifinals**, selected among **11 teams** from **88 competitors** and competing for **$50,000** in startup funding
- Conducted **7** clinical stakeholder interviews with physicians, researchers, and healthcare leaders to validate unmet clinical needs and translate user feedback into device requirements and product strategy

Zimmer Biomet **Montreal, QC**
Systems Design Intern May 2025 - August 2025
- Tested the ROSA Knee System by performing statistical analysis in **Minitab** to evaluate biomechanical cut accuracy as part of the performance team ensuring compliance with **ISO 13485** QMS and design validation requirements
- Reduced processing from **3** minutes to **3** seconds across **200+** executions by developing a **SolidWorks** macro that automated alignment of femur and tibia models using 3D spatial coordinate algorithms for precise validation of robot calculated cut planes
- Authored sections of **7 FDA compliant DVeP & DVeRs** for medical device design, gaining experience in regulatory compliant documentation and traceability processes in a french speaking engineering environment

Insead Business School for the World **Fontainebleau, France**
Chief Operating Officer during Summer Program July 2022 - August 2022
- Led as the Chief Operating Officer of a **6** member team in a simulated bicycle company, streamlining manufacturing by reducing design cost by **30%** and implementing **risk analysis** and mitigation strategies to ensure product safety
- Established bold manufacturing and rebranding changes to differentiate ourselves from competitors and drive innovation

Projects

Computational Modeling of Action Potentials **Los Angeles, CA**
Student Researcher September 2025 - November 2025
- Characterized system behavior across **50+** simulations by developing an **Excel** and **MATLAB** based computational models of neuronal action potentials, analyzing voltage propagation, spike frequency, and conductance behavior under varying excitation and inhibition inputs to investigate system performance and variability

Electric Guitar **Los Angeles, CA**
Electrical Design Lead March 2025 - May 2025
- Developed and simulated an active filter circuit in **LTspice** utilizing high and low-pass op-amp filters for analog signal conditioning and tonal control
- Designed **KiCad** schematics and **PCB** layouts, soldering, and integrating the electronics into a functional electric guitar

Smart Knee Brace **Los Angeles, CA**
CAD Design Lead January 2025 - February 2025
- Constructed an **Arduino**-powered smart knee brace utilizing **Fusion360** to assist athletes in recovering from patellar tendonitis **2x** faster by integrating vibration therapy and real-time planar knee motion-tracking using accelerometers
- Implemented haptic feedback alerts to notify users of risky movements, promoting safer rehabilitation

Skills & Extracurriculars

Technical Skills: Solidworks, Fusion360, Siemens NX, AutoCAD, Autodesk Fusion, PTC Creo, CATIA V5, ICIDO, ENOVIA, Revit, TinkerCAD, MATLAB, Python, C/C++, Java, Arduino IDE, Labview, KiCad, LTspice, Microsoft Excel, MS Office, Minitab, DFMEA, RCA, GMP, QA, ISO 13485 QMS, GD&T, FEA, DFMA, 3D Printing, PCR, Digital signal processing, Soldering, Welding, Oscilloscope, Strain gauge, Load cell, Accelerometer, Thermocouple, Pitot tube, FTIR, Gyroscope
Languages: Fully Bilingual in English and French
Organizations: Theta Tau Professional Engineering Society, Engineers Without Borders, MEDesign, Ski Team, Brazilian Jiu Jitsu
International Exposure: US and French Citizen, raised in Singapore; studied/worked in the U.S., Canada, France, and Czechia